

Decision memo: implications of the multi-proxy human-impact model for H1

Internal discussion document for Suzette, Vivian, and Ondřej

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Decision requested. Pause the current paper-wide statement that climate outweighs human impact. Review the harmonised H1 comparison below, agree what construct should represent human impact, and decide whether H2 must be refitted before the manuscript narrative is revised.

1 Executive summary

The new result is consequential, but it is interpretable rather than chaotic. In the spatial H1 analysis, the canonical SPD model is climate-dominant for 869 of 1,069 common sequences, whereas the filtered joint-human model is human-dominant for 692 sequences and climate-dominant for 376. The median zero-truncated human-minus-climate balance moves from -0.53 to 0.18. This is a spatial reversal.

The temporal result weakens rather than fully reverses. Climate exceeds humans in 47 of 65 continental-region-age units under the joint model, compared with 64 of 65 canonically. Median human allocation rises from 5% to 26%, while median climate allocation falls from 70% to 49%.

The matched SPD bridge is decisive for interpretation. It uses the same 8–2 ka observations and exactly the same locally retained climate variables as the joint model, but retains square-root SPD as the only human predictor. The bridge remains climate-dominant spatially (764 versus 305) and in all 65 temporal units. Moving from the bridge to the joint human block changes 387 spatial sequences and 18 temporal units from climate- to human-dominant, with no corresponding human-to-climate reversals. The expanded human representation is therefore the main source of the changed conclusion on the harmonised cohort.

H2 has not been harmonised: it remains SPD-only. Consequently, neither the canonical climate-dominance headline nor a new paper-wide human-dominance headline is currently supported across both hypotheses.

2 What changed methodologically

The revised analysis treats human impact as a multivariate construct, closer in structure to the multivariate climate block. Square-root SPD represents archaeological activity or demography; KK10 represents land-use fraction; and square-root HYDE represents population. Square-root transformations reduce the leverage of the long right tails of SPD and HYDE without changing their ordering.

Within each pollen sequence or continental-region-age unit, redundant variables are removed *within* the human and climate blocks at $|r| = 0.8$ and $VIF = 5$. Selection has no access to the response. Time and selected dbMEM variables remain protected controls, and cross-block correlations are diagnosed rather than used for automatic removal because their shared variation is the object being partitioned.

Table 1: The bridge-to-joint contrast changes only the human block.

Specification	Human block	Climate block	Interval / cohort	Interpretive role
Canonical paper	$\sqrt{\text{SPD}}$	Four fixed climate predictors	8.5–2 ka; canonical cohort	Current manuscript result
Matched SPD bridge	$\sqrt{\text{SPD}}$	Locally filtered; identical to joint	8–2 ka; matched cohort	Separates interval, cohort, and climate-filter changes
Filtered joint	$\sqrt{\text{SPD}} + \text{KK10} + \sqrt{\text{HYDE}}$, locally filtered	Locally filtered; identical to bridge	8–2 ka; matched cohort	Tests the expanded human construct

The climate-set equality assertion passed for all 1175 sequence-level units and all 65 continental-region-age units. The pipeline reported no unclassified model errors. All comparisons below use the exact three-model common cohort: 1,069 sequences for the spatial pattern and 65 continental-region-age units for the temporal pattern.

The canonical-to-joint contrast remains useful as a manuscript sensitivity comparison, but it combines changes in interval, cohort, climate selection, and human representation. Only the bridge-to-joint contrast isolates expansion of the human block.

3 Figure 3 comparison: spatial pattern

These are direct Figure-3-style outputs using the same 1,069 sequences, map limits, balance scale, region ordering, spatial-control workflow, and zero-truncated composition. The only intended differences are those shown in the specification table.

A. Canonical SPD model. Climate dominates most sequence-level balances and regional distributions.

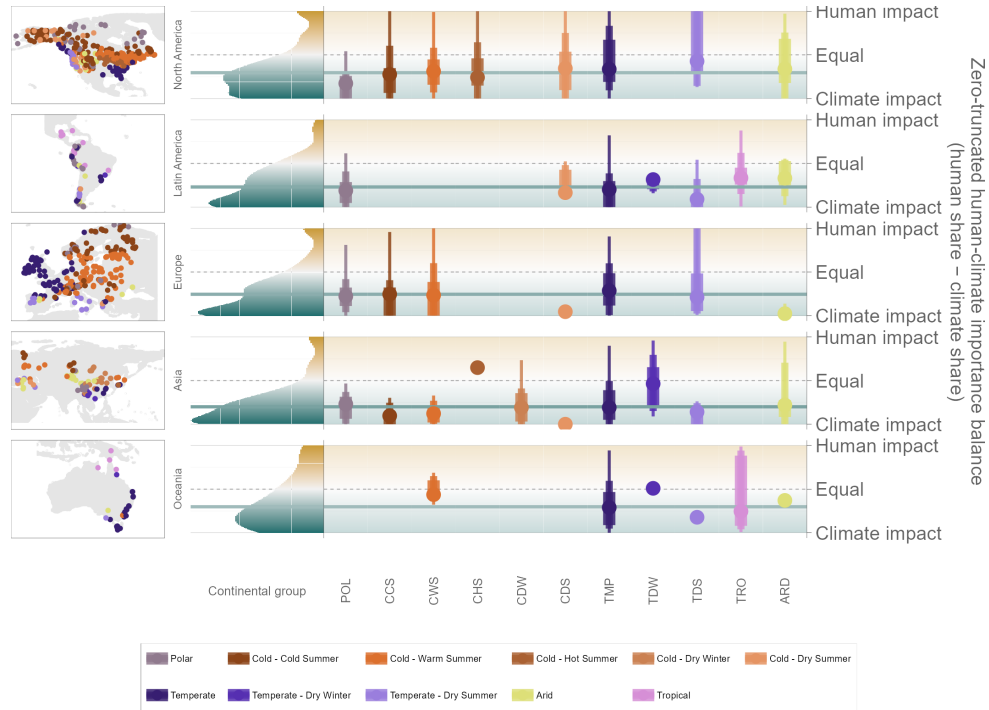


Figure 1: Figure 3 variant A. Canonical square-root SPD model on the exact three-model common cohort.

B. Matched SPD bridge. Matching the interval and cohort and filtering climate moves the distribution towards zero, but the overall pattern remains climate-dominant.

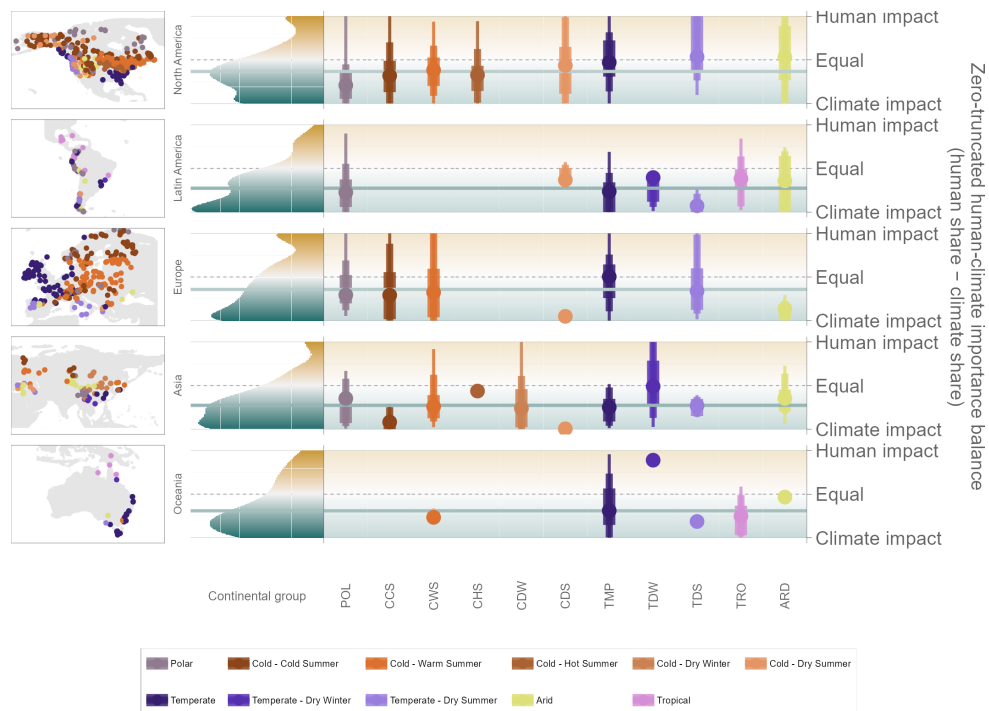


Figure 2: Figure 3 variant B. Matched square-root SPD bridge with the joint model's locally selected climate predictors.

4 Figure 3 comparison: joint human block

C. Filtered joint-human model. The spatial pattern changes direction when the human block can retain square-root SPD, KK10, and square-root HYDE.

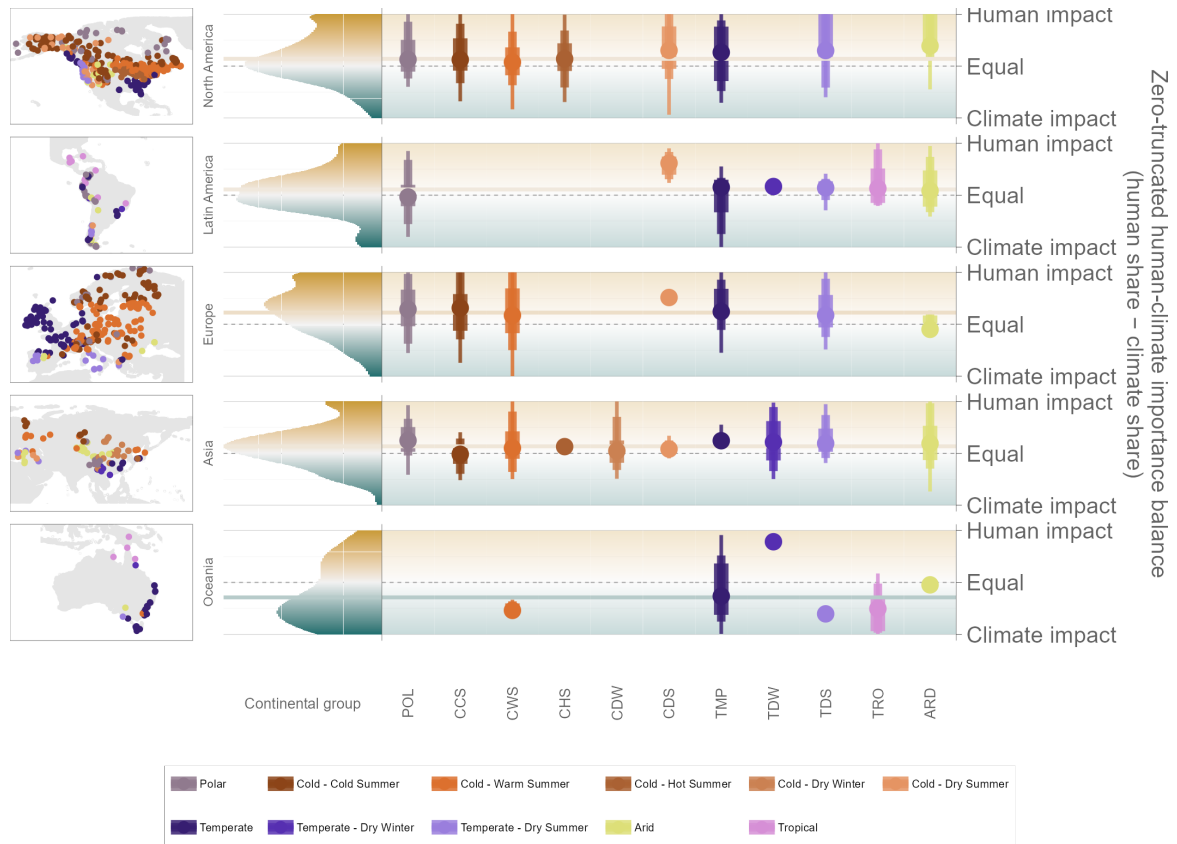


Figure 3: Figure 3 variant C. Filtered joint-human model on the same exact common cohort.

Table 2: Spatial H1 summary on the exact common cohort.

Model	Human larger	Climate larger	Median balance	Median total adj. R2
Canonical SPD	200	869	-0.53	0.453
Matched SPD bridge	305	764	-0.35	0.429
Filtered joint human block	692	376	0.18	0.483

The bridge remains climate-dominant, whereas adding the filtered KK10 and HYDE information creates 387 climate-to-human reversals and moves the median balance from -0.35 to 0.18.

5 Figure 4 comparison: temporal pattern

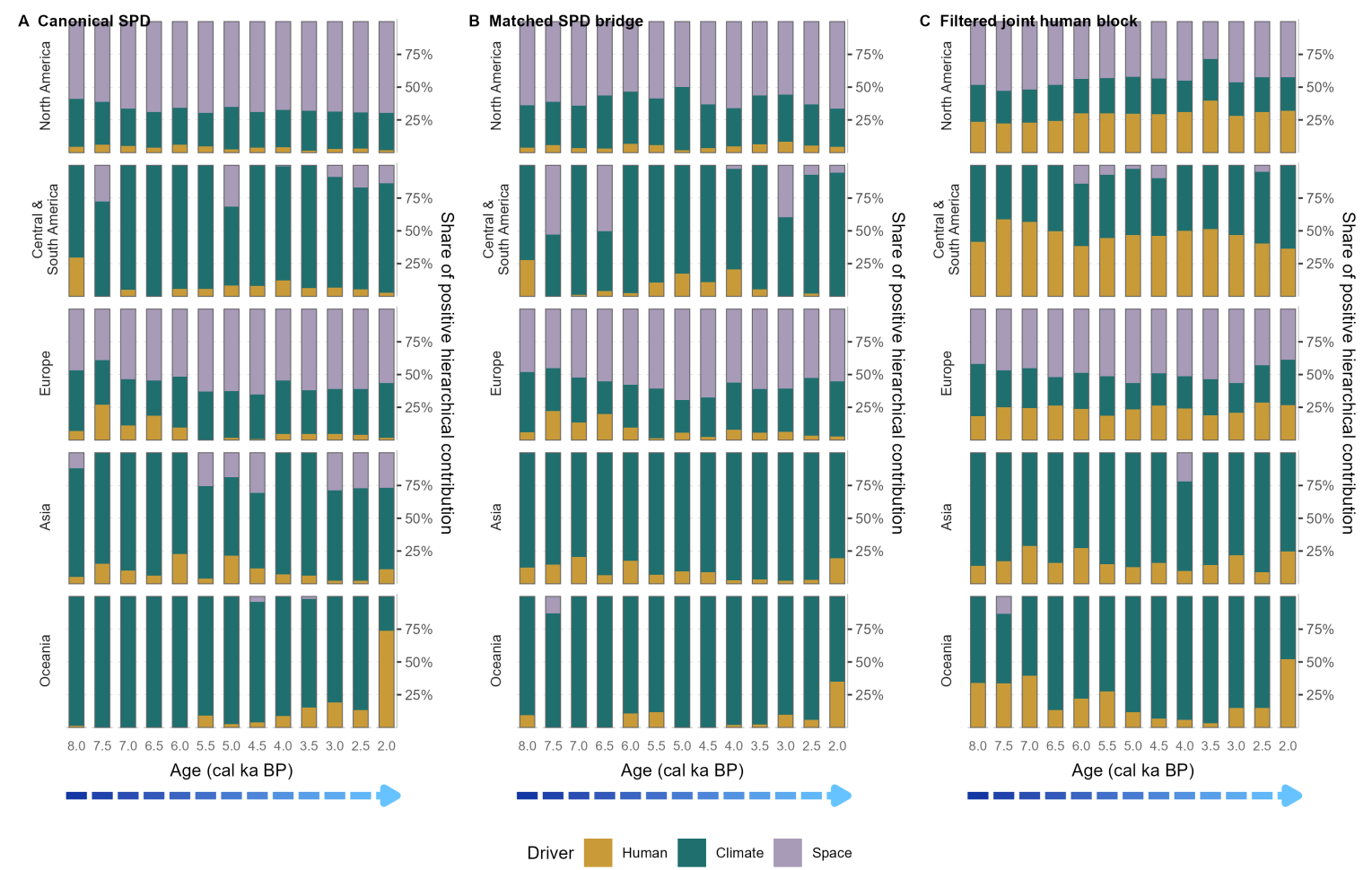


Figure 4: Figure 4 variants on the same 65 continental-region-age units: canonical SPD, matched SPD bridge, and the filtered joint-human block.

Climate exceeds humans in 64, 65, and 47 of 65 for the canonical, bridge, and joint variants, respectively. Thus the temporal result weakens substantially under the joint block but does not reverse overall.

6 Interpretation and manuscript choices

The joint model should not automatically be called “correct”. Important limitations remain:

- block dimensionality can affect how much variation a block can capture, even after filtering;
- local selection makes the meaning of the human and climate blocks vary among units;
- cross-block shared variation is scientifically relevant but prevents simple causal attribution;
- time and space controls can themselves be strongly related to focal predictors;
- negative adjusted components and suppression make individual fractions unstable in some units; and
- zero truncation produces an interpretable composition but conceals negative fractions.

The robust statement is therefore narrower: **the claim that climate generally outweighs humans in H1 is not robust to defensible representation of human impact**. Spatial H1 reverses; temporal H1 remains climate-dominant overall but becomes substantially less uniform.

6.1 Three manuscript options

Table 3: Consequences of the three defensible manuscript strategies.

Option	Title_and_abstract	H1_results_and_figures	Discussion_and_review	H2_consequence
1. Keep SPD primary; soften headline	Remove broad climate-dominance claim; frame SPD explicitly as archaeological activity.	Keep current main figures; add joint sensitivity prominently and report non-robustness.	Lowest structural cost, but must explain why a narrower human construct remains primary.	Internally consistent with current SPD-only H2, but construct remains narrow.
2. Replace primary H1 with joint model	Reframe human impact as a multivariate construct; spatial and temporal conclusions must be separated.	Replace main H1 figures and rewrite results around spatial reversal and temporal weakening.	Strong methodological symmetry, but largest revision and strongest need for additional validation.	Requires a joint-proxy H2 refit before any paper-wide conclusion.
3. Treat SPD and joint models as co-primary	Lead with representation sensitivity or scale dependence rather than one dominant driver.	Present matched SPD bridge and joint model together; distinguish robust from specification-dependent patterns.	Most transparent about uncertainty, but increases narrative and figure complexity.	H2 should be harmonised to avoid asymmetric evidence across hypotheses.

My process recommendation is not to select an option in this memo. It is to suspend the existing headline, discuss what construct the team intends “human impact” to mean, and decide whether the H2 refit is required before manuscript restructuring.

7 Proposed H2 refit

Before adopting a paper-wide joint-proxy narrative, H2 should be brought onto the same evidential footing:

1. construct regional KK10 and HYDE trajectories on the same age grid and spatial units used by the existing H2 analysis;
2. preserve square-root SPD, KK10 as a fraction, and square-root HYDE;
3. apply response-independent local correlation/VIF filtering within the human and climate blocks;
4. fit a harmonised SPD-only bridge and a filtered joint-human model on an exact common cohort;
5. preserve the existing H2 response, controls, HVarPart logic, thresholds, and diagnostics; and
6. report exclusions, selected predictors, total and unique adjusted R^2 , negative components, and residual dependence for both models.

This is not a request to make H2 agree with H1. It is a request to remove the current asymmetry in how “human impact” is represented before drawing a paper-wide conclusion.

8 Questions for the co-author meeting

1. Which construct best represents “human impact” for the primary hypothesis: archaeological activity alone, or a combined demography–land-use–population construct?
2. Is a scale-dependent conclusion — spatial reversal but temporal weakening — scientifically preferable to any single dominant-driver claim?
3. Should H2 be rerun before we choose a manuscript structure, or can we first agree the conceptual model?
4. Which of the three manuscript options is most defensible to readers and reviewers, even if it requires the most rewriting?
5. What additional robustness check would materially change our decision, rather than only add another sensitivity panel?

Suggested meeting outcome. Agree the intended human-impact construct, select a provisional manuscript option, and decide whether to commission the harmonised H2 refit before changing the title, abstract, or main figures.

9 Technical appendix: scope and provenance

This memo is an internal decision aid, not a reviewer response and not manuscript text. It does not modify the canonical H1 or H2 pipelines, their target stores, the manuscript, or the submitted figures. The matched bridge and joint models run in the isolated target store `sensitivity_analyses/human_proxy_hvarpart_collinearity_managed`.

The evidence covers matched observations between 8,000 and 2,000 cal yr BP without extrapolation. The spatial comparison is restricted to the 1,069 sequences eligible under all three specifications. The temporal comparison includes all 65 common continental-region-age units. Every temporal composition was checked to sum to one after zero truncation, the bridge and joint climate selections were asserted to be identical within every analytical unit, and no caught model error remained unclassified.

Check	Result
Spatial common cohort	1,069 sequences
Temporal common cohort	65 continental-region-age units
Bridge/joint climate selections	Identical in every unit
Temporal stack reconstruction	All sums equal one within numerical tolerance
Unclassified model errors	None
Primary interpretation	Bridge-to-joint isolates expansion of the human block
Scope restriction	H1 only; H2 remains SPD-only

Primary evidence files are the exported `decision_spatial_*`, `decision_temporal_*`, `shared_climate_selection.csv`, adjusted- R^2 , and predictor-selection tables under `Outputs/Tables/H1/Sensitivity/Human_proxy_hvarpart_collinearity`. The QMD stops with the exact pipeline command if these files are missing, and all analytical numbers in this memo are computed from those public CSVs at render time.

Bottom line: the original headline is no longer safe as a general conclusion. The strongest next step is a team decision about the human-impact construct, followed by a harmonised H2 refit if the joint construct is to shape the paper-wide narrative.